
INTERNSHIP INTERVIEW EXPERIENCE

ABOUT THE COMPANY:

Vegrow, India's leading B2B fruits marketplace, was founded in 2020 by IIT alumni. The company has established a strong presence in India's large, unorganized fruits sector, which is ripe for tech-driven disruption due to its fragmented and localized operations, characterized by multiple intermediaries facilitating transactions. Vegrow maximizes farmers' income by accurately grading produce and efficiently matching it with the most suitable demand channels. This creates a virtuous cycle, attracting more farmers through competitive pricing, as well as more buyers who seek consistency in both price and quality.

INTERVIEW EXPERIENCE:

ROUND 1 (Coding):

The round 1 is basically an offline coding contest conducted in a hacker rank portal comprising of four different questions.

0/1 Knapsack Problem:

This is a classic Dynamic programming problem where you have to pick items within a weight limit with max profit that can be achievable. This problem is a must to know (both the recursive and optimised approach). To know more: <https://www.geeksforgeeks.org/0-1-knapsack-problem-dp-10/>

SQL Query:

A SQL query to retrieve students` name whose marks is higher than 75 sorted by the lexicographical order of their last three letters of their names if they were similar then it should be further sorted using their id. Hint: use aggregate functions such as substr ().

Best time to buy and sell stock:

In this problem, an array with value of the stocks in a day will be given, our task is to find the maximum profit we can achieve by buying and selling a stock. For reference:

<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/description/>

Reward Distribution Problem:

An array will be given where $\text{array}[i]$ represents the productivity rating of a field i . Our task is to assign rewards to these fields such that the field with more productivity rating than the adjacent fields must have higher rewards.

TECHNICAL INTERVIEW

ROUND 1:

First and foremost, I was asked to introduce myself, during which I highlighted my achievements and interests. I mentioned my passion for Data Structures and Algorithms (DSA) and stated that I have solved over 250 problems on LeetCode. They then asked for my LeetCode profile and inquired about my favourite data structure. I told them that I love trees. Following that, they asked me a question similar to the [house robber](#) on LeetCode.

Note: Even if you know the optimized approach to a problem, it's best to start by explaining the brute-force method first.

I wrote pseudocode for the problem, and while there were some logical mistakes, the solution was acceptable to them. Interviewers often focus on how you approach and think through a problem rather than on providing a completely flawless solution.

Next, they gave me their laptop and asked me to solve the [problem](#). I managed to find an approach that worked. Then, they asked about the SQL query from the coding round and gave me a modified query problem to solve. They also asked whether I was proficient in the Standard Template Library (STL), to which I responded that I was comfortable with it.

They inquired about the difference between unordered map and map, and I explained the complexity of both in terms of insertion and how they are implemented. Lastly, they asked me a few questions related to my resume. As a final challenge, they expected me to solve a [problem](#) with the most optimized approach in terms of both time and space complexity.

ROUND 2:

For this round, five of us were shortlisted. First, they asked me about the projects on my resume. While I was explaining one of my projects, which involved working with a map, they posed some questions about the process of finding the shortest path between two points.

After that, they asked me two DSA questions:

1. You are given a string, and the task is to remove characters that have adjacent similar elements. We have to continue removing characters as long as they share adjacent similarities, even after removal. (Hint: Use a Stack)
2. They also asked me to write pseudocode for printing the nodes that are visible in the bottom view of a tree. You can refer to this [Bottom View of Binary Tree problem](#).

Finally, an aptitude-based puzzle question was asked, which you can refer to here: [Puzzle 18: Torch and Bridge](#).

RESOURCES

1. <https://github.com/singhsanket143/Data-Structures-Algorithms-Problem-Solving/blob/master/README.md>
2. <https://github.com/Chanda-Abdul/Several-Coding-Patterns-for-Solving-Data-Structures-and-Algorithms-Problems-during-Interviews/blob/main/%E2%9C%85%20%20Pattern%2001%20%3A%20Sliding%20Window.md>
3. https://docs.google.com/document/d/1jfaa6loWq4oJ7zGv_np8HO84HaB0OIhEapQzSS-c5l4/edit
4. <https://takeuforward.org/interviews/strivers-sde-sheet-top-coding-interview-problems/>
5. https://www.youtube.com/watch?v=0bHoB32fuj0&list=PLgUwDviBIf0oF6QL8m22w1hIDC1vJ_BHz

TIPS TO LEARN SMART

1. Focus on the problem-solving patterns like sliding window, two pointers, etc., rather than solving the individual problem. There's always a pattern in every problem waiting to be found. So, focus on finding the strategy and understand the strategy.
2. It is always the case that you feel often demotivated and stagnated in the same level but really here consistency matters. Noone knows the right and optimised solution for a problem right away. It really takes a vast amount of time to build the intuition. So don't be frustrated, see and learn from the answer and implement it by yourself which is the best strategy to learn and grow.
3. If you have done solving the problem look into the top submissions, there's always lot to learn from the top submissions. It really holds the key to improve your code.
4. Start with simple data structures like arrays and linked list and climb up into graph and trie, etc..., Most of the companies never bother about very hard data structures and algorithm.
5. Always express your thought process to the interviewer. The interviewer looks for the thought process while solving a problem rather than a complete solution to the problem. Even if you are not able to solve the problem just convey your thoughts.
6. Finally have insane luck even though most of the above points matter. The luck really determines where you would really end up. ~~(It's hard but it's the reality thou)~~.

ALL THE BEST

From,

VIJAY K G

2022115063

B. TECH IT

Batch of 2026

Reach me at 8438938916